

Tutorial 02 – Testing Metrics

TG/2022/1388

1. Requirement Metrics

Requirement Creep

$$\begin{aligned} &= (\text{Added Requirements} / \text{Initial Requirements}) \times 100 \\ &= (18 / 120) \times 100 \\ &= \mathbf{15\%} \end{aligned}$$

Requirement Coverage

$$\begin{aligned} &= (\text{Covered Requirements} / \text{Total Requirements}) \times 100 \\ &= (280 / 138) \times 100 \\ &= \mathbf{202.899\%} \end{aligned}$$

2. Defect Metrics

Defect Density

$$\begin{aligned} &= \text{Total Defects} / \text{KLOC} \\ &= 160 / 10 \\ &= \mathbf{16 \text{ defects per KLOC}} \end{aligned}$$

Defect Leakage

$$\begin{aligned} &= (\text{Post-release Defects} / \text{Total Defects}) \times 100 \\ &= (20 / 160) \times 100 \\ &= \mathbf{12.5\%} \end{aligned}$$

Defect Removal Efficiency (DRE)

$$\begin{aligned} &= (\text{Defects Found before release} / \text{Total Defects}) \times 100 \\ &= (110 / 160) \times 100 \\ &= \mathbf{68.75\%} \end{aligned}$$

Critical Defects Percentage

$$\begin{aligned} &= (\text{Critical Defects} / \text{Total Defects}) \times 100 \\ &= (35 / 160) \times 100 \\ &= \mathbf{21.88\%} \end{aligned}$$

3. Test Case Metrics

Passed Percentage

$$\begin{aligned} &= (\text{Passed Test Cases} / \text{Executed Test Cases}) \times 100 \\ &= (210 / 280) \times 100 \\ &= \mathbf{75\%} \end{aligned}$$

Failed Percentage

$$\begin{aligned} &= (\text{Failed Test Cases} / \text{Executed Test Cases}) \times 100 \\ &= (50 / 280) \times 100 \\ &= \mathbf{17.86\%} \end{aligned}$$

Blocked Percentage

$$\begin{aligned} &= (\text{Blocked Test Cases} / \text{Executed Test Cases}) \times 100 \\ &= (20 / 280) \times 100 \\ &= \mathbf{7.14\%} \end{aligned}$$

4. Effort and Cost Metrics

Rework Effort Ratio

$$\begin{aligned} &= (\text{Rework Effort} / \text{Total Testing Effort}) \times 100 \\ &= (40 / 200) \times 100 \\ &= \mathbf{20\%} \end{aligned}$$

Cost of Finding a Defect

$$\begin{aligned} &= \text{Testing Effort} / \text{Defects found} \\ &= 200 / 110 \\ &= \mathbf{1.82 \text{ hours per defect}} \end{aligned}$$

Average Repair Time

$$\begin{aligned} &= \text{Total Bug Fix Time} / \text{Number of bugs} \\ &= 180 / 130 \\ &= \mathbf{1.38 \text{ hours per defect}} \end{aligned}$$

5. Productivity Metrics

Test Design Efficiency

= Test Cases Designed / Design Time

= 300 / 15

= **20 test cases per hour**

Test Execution Rate

= Test Cases Executed / Testing Time

= 280 / 200

= **1.4 test cases per hour**

Developer to Bugs Ratio

= Total Defects / Number of Developers

= 160 / 6

= **26.67 bugs per developer**